

A205 · SIGMA: interface-aware spectral graph learning for metabolic transitions across pathological tissue boundaries

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Session: Day 2 · Fri Oct 2, 2026 · 2:15–4:15 PM

Pathological annotations often capture only visible lesions, whereas molecular alterations can extend into surrounding tissue. Here we present SIGMA, an interface-aware spectral graph method integrating spatial metabolomics, tissue coordinates and, where available, spatial transcriptomics and pathological anchors to resolve continuous metabolic transitions. SIGMA separates broad tissue organisation from local interface-associated variation and quantifies signed distance, influence range, transition sharpness and directional anisotropy. In simulations, SIGMA distinguished interface-associated features from regional and generic spatial patterns. Across five cancers, SIGMA identified metabolic programmes under direct pathological annotations, transcriptomics-informed weak anchors, transferred annotations and metabolomics-only settings. Cross-cancer analysis revealed continuous, overlapping interface phenotypes rather than discrete classes. Matched spatial transcriptomics further supported feature prioritisation and biological interpretation. Beyond cancer, SIGMA captured metabolic organisation across a broad dopamine-associated transition in Parkinson's disease striatum. SIGMA thus provides a quantitative approach for mapping metabolic transitions beyond conventionally annotated pathological boundaries.